BCSE204L Design and Analysis of Algorithms

Module-4: Topic-1:Bellman Ford Algorithm

Single Source Shortest Path-Bellman-Ford Algorithm

- Dijkstras algorithm is not guaranteed to work for graphs with negative weighted edges.
- Dynamic programming approach
- Try out all possibilities and select the best one
- Process: Relaxation

- To be done all vertices.
- Maximum $|V| - 1$ edges between any two vertices. Relaxation

$$
\begin{aligned}
\text{For any edge } (u, v) \\
\text{if } d[u] + c(u, v) < d[v] \\
\text{then } d[v] = d[u] + c(u, v)
\end{aligned}
$$

Bellman Ford Algorithm- Demo

On board

Bellman Ford Algorithm- Drawback

On virtual board

```
function bellmanFord(G, S)
  for each vertex V in G
    distance[V] <- infinite
      previous[V] <- NULL
  distance[S] <- 0

  for each vertex V in G
    for each edge (U,V) in G
      tempDistance <- distance[U] + edge_weight(U, V
      if tempDistance < distance[V]
        distance[V] <- tempDistance
        previous[V] <- U
```

Algorithm Cont..

```
for each edge (U,V) in G
  If distance[U] + edge_weight(U, V) < distance[V]
   Error: Negative Cycle Exists
return distance[], previous[]
```

- $|E|$ edges are relaxed $|V| - 1$ times
  $\Rightarrow O(|E||V|)$
- Complete graph of n vertices has
  $n(n-1)/2$ edges relaxed for n times
  $\Rightarrow O(n^3)$

Thank You